Week 4

DevOps Internship

Release Date: 17 August 2026

This week, you'll learn **Docker & Docker Compose**. Containerization makes applications portable, consistent, and easy to deploy across different environments, making Docker one of the most essential tools in modern DevOps.

Topic:- Docker & Docker Compose

This week, you are expected to learn:

- Introduction to Docker
- Docker Architecture
- Docker Installation
- Docker Images, Docker Containers
- Docker Hub
- Dockerfile
- Docker Volumes
- Docker Networks
- Environment Variables
- Port Mapping
- Docker Compose
- Multi-Container Applications
- Docker Best Practices

***Learning Approach**

To get the most out of this week's learning, focus on understanding how containers work instead of simply running Docker commands. Practice building, running, and managing containers while understanding the purpose of each Docker component.

Keep these points in mind:

- ✓ Build Docker images from scratch using a Dockerfile.
- ✓ Understand the difference between Images, Containers, Volumes, and Networks.
- ✓ Practice Docker commands instead of memorizing them.
- ✓ Deploy multi-container applications using Docker Compose.
- ✓ Verify AI-generated Docker configurations before using them.

Weekly Task

***Complete the following practical exercises:**

- Install Docker Desktop or Docker Engine.
- Verify the Docker installation.
- Pull images from Docker Hub.
- Create and manage Docker containers.
- Build your own Docker image using a Dockerfile.
- Run a simple HTML or Node.js application inside a container.
- Configure Docker Volumes for persistent storage.
- Create a custom Docker Network.
- Deploy a multi-container application using Docker Compose.
- Push your Docker image to Docker Hub.
- Prepare a short PDF explaining:
    - What is Docker?
    - Docker Architecture
    - Images vs Containers
    - Dockerfile
    - Docker Volumes
    - Docker Networks
    - Docker Compose
    - Docker Hub

Task Submission

Once you complete the above task, submit:

- GitHub Repository Link
- PDF Report
- Dockerfile
- Docker Compose File
- Screenshot of Docker Installation
- Screenshot of Running Containers
- Screenshot of Docker Images
- Screenshot of Docker Hub Repository

Task Submission Deadline: Wednesday (EOD)

Hands-on Activity

Scenario

Imagine you have joined a company as a **Junior DevOps Engineer**. Your manager asks you to containerize a web application and deploy it using Docker Compose for a consistent development environment.

Complete the following:

- Create a Dockerfile for a simple HTML or Node.js application.
- Build a custom Docker image.
- Push the image to Docker Hub.
- Create a Docker Compose file.
- Deploy a multi-container application consisting of:
  - Frontend
  - Backend
  - Database
- Configure Docker Volumes for persistent storage.
- Configure a custom Docker Network.
- Verify that all containers are communicating successfully.
- Document every implementation step.

Activity Submission

After completing the hands-on activity, submit:

- ✓ GitHub Repository Link
- ✓ Dockerfile
- ✓ Docker Compose File
- ✓ Screenshot of Running Containers
- ✓ Screenshot of Docker Hub Repository
- ✓ Screenshot of Running Application
- ✓ Short implementation summary

Activity Submission Deadline: Friday (EOD)

AI Assistant Task

Use an AI Assistant such as **ChatGPT, GitHub Copilot, Amazon Q, Gemini, Claude, or Microsoft Copilot** to strengthen your understanding of Docker.

***Complete the following:**

- Ask AI to generate a Dockerfile for your application.
- Ask AI to explain Docker Images, Containers, Volumes, and Networks.
- Generate a Docker command cheat sheet with at least **25 commonly used Docker commands**.
- Ask AI to create a Docker Compose file for a multi-container application.
- Generate a Docker troubleshooting scenario and resolve it.
- Verify the AI-generated configuration by testing it in your Docker environment.

DevOps Pro Tip

Keep your Docker images lightweight by using minimal base images and excluding unnecessary files with a dockerignore file. Smaller images build faster, deploy faster, and improve overall application performance.

Take time to understand how Docker containers work instead of simply running Docker commands. Building a strong foundation in containerization will make Kubernetes, CI/CD pipelines, and cloud-native deployments much easier throughout your DevOps journey.

If you face any genuine issues, feel free to reach out before the submission deadline.